A-A-51130A
February 3, 1990
SUPERSEDING
A-A-51130
August 14, 1984

COMMERCIAL ITEM DESCRIPTION

CUTTER BIT BLANKS AND CUT-OFF BLADES

The General Services Administration has authorized the use of this commercial item description.

1. SCOPE

1.1 Scope. This description covers cutter bit blanks and cut-off blades of high speed steel that are supported by tool holders for use in performing various metalworking machining operations.

1.2 Classification. The cutter bit blanks and cut-off blades shall be of the following types, styles, grades, classes, and sizes. The types, styles, grades, classes, and sizes to be furnished shall be as specified (see 6.1).

Type I - Cutter bit blanks.
  Style 1 - Square cross-section (see table I).
  Style 2 - Rectangular cross-section (see table II).

Type II - Cut-off blades.
  Style 1 - Straight side clearance (see table III).
  Style 2 - Armstrong style (see table IV).

Steel grade A - General purpose.
  Class - M-1, M-2, M-3, M-4, or M-10.

Steel grade B - Heavy duty.
  Class - T-15, M-30, M-34, M-41, M-42, M-43, M-44, or 46.

Sizes. The sizes and dimensions of the cutter bit blanks and cut-off blades shall conform to tables I, II, III, and IV as required for the type, style, grade, class, and size item specified (see 6.1). The tolerances for all dimensions shall be manufacturer's standard as specified on their drawings.

AMSC N/A FSC 3455
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TABLE I. Type I, style 1, square cross-section, cutter bit blanks.

Size	Dimensions in inches	
	Square cross-section	Length
00	3/16	2
01	1/4	2-1/2
03	5/16	2-1/2
04	3/8	3
05	7/16	3-1/2
06	1/2	4
07	5/8	4-1/2
08	3/4	5
09	7/8	6
10	1	7
12	1-1/8	8
13	1-1/4	9
14	1-1/2	10

TABLE II. Type I, style 2, rectangular cross-section, cutter bit blanks.

Size	Dimensions in inches		
	Width	Height	Length
15	1/4	3/8	2-1/2
16	1/4	1/2	4
17	1/4	1/2	6
18	3/8	1/2	3
19	3/8	1/2	4
20	3/8	1/2	6
21	3/8	5/8	6
22	3/8	3/4	6
23	1/2	3/4	4
24	1/2	3/4	6
25	1/2	1	6
26	5/8	7/8	6
27	3/4	1	6
28	1	1-1/2	10
29	1	2	12

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TABLE III. Type II, style 1, straight side clearance, cut-off blades.

Size	Dimensions in inches		
	Thickness	Height	Length
30	1/16	13/32	3
31	1/16	1/2	4-1/2
32	1/16	11/16	5
33	1/16	13/16	6
34	5/64	1/2	4-1/2
35	3/32	1/2	4-1/2
36	3/32	5/8	5
37	3/32	13/16	6
38	3/32	1	6-1/2
39	1/8	1/2	4-1/2
40	1/8	3/4	5
41	1/8	7/8	6
42	1/8	1	6-1/2
43	5/32	11/16	5
44	5/32	13/16	6
45	3/16	11/16	5
46	3/16	13/16	6
47	3/16	7/8	6
48	3/16	1	6-1/2
49	3/16	1-1/8	7
50	1/4	13/16	6
51	1/4	7/8	6
52	1/4	1	6-1/2
53	1/4	1-1/8	6-1/2
54	1/4	1-1/4	7

TABLE IV. Type II, style 2, armstrong style, cut-off blades.

Size	Dimensions in inches			Top flat width	
	Thickness	Height	Length	Minimum	Maximum
60	3/32	1/2	4-1/2	0.046	0.063
61	3/32	5/8	5	0.046	0.063
62	1/8	3/4	5	0.062	0.081
63	1/8	7/8	6	0.062	0.081
64	3/16	1	6-1/2	0.093	0.114
65	3/16	1-1/8	7	0.093	0.114
66	1/4	1-1/4	7	0.141	0.154
67	1/4	1-3/8	7	0.141	0.154

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2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of the issue in effect on the date of invitation for bids or request for proposals form a part of this description to the extent specified herein.

Federal Specification PPP-P-40	- Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven Metal and Wood-working Machinery.
Federal Standard FED-STD-376	- Preferred Metric Units for General Use by the Government.
American National Standard Institute ANSI B94.3	- Straight Cut-off Blades for Lathes and Screw Machines.
American Society for Testing and Materials ASTM D 3951	- Standard Practice for Commercial Packaging.
American Society for Testing and Materials ASTM A-600	- Standard Specification for High Speed Tool Steel.

3. SALIENT CHARACTERISTICS

3.1 Design. The cutter bit blank and cut-off blade shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The cutter bit blank and cut-off blade shall meet the requirements specified herein and conform to current industrial practice.

3.2 Materials. Cutter bit blank and cut-off blade shall be high-speed tool steel alloy, hardened and tempered to a Rockwell "C" hardness of not less than 63 with the grades and classes conforming to the requirements of ASTM A 600.

3.3 Type I cutter bit blanks. Type I, style 1, cutter bit blanks shall have a square cross-section with ends beveled to each other at an angle of 20 degrees and the type I, style 2, blanks shall have a rectangular cross-section with the ends squared.

3.4 Type II cut-off blades. Type II, style 1, cut-off blades shall conform to ANSI B94.3. Type II, style 2, blades shall conform to ANSI B94.3, Type "F".

3.5 Measurement system. All dimensions in this description are given in the inch/pound system. These dimensions may be converted to SI system through the use of conversion factors and methods specified in FED-STD-376.

3.6 Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

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3.7 Workmanship. Workmanship of the cutter bit blanks and cut-off blades shall be of a quality equal to that of comparable products furnished to the commercial market.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this description shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptance practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Sampling inspection. As a minimum sampling inspection, the supplier shall randomly select the number of samples in table V that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination in 4.3 and the dimensional inspectionin 4.4. Should any of the samples fail to meet either the examination or inspection below, the entire lot shall be considered defective and unsuitable for acceptance. A lot is defined as all items of the same type, style, size, and class presented at one time for acceptance.

TABLE V. Sampling plan.

Quantity of items being purchased	Sample size	Quantity of items being purchased	Sample size
2 to 8	2	91 to 150	20
9 to 15	3	151 to 280	32
16 to 25	5	281 to 500	50
26 to 50	8	501 to 1200	80
51 to 90	13		

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4.3 Examination. Each cutter bit blank and cut-off blade submitted for sample inspection shall be visually examined to determine conformance with all requirements of this description except the hardness tests.

4.4 Dimensional inspection. Each cutter bit blank and cut-off blade submitted for sample inspection shall be inspected to assure conformance with all dimensional and tolerance requirements specified herein.

4.5 Hardness test. After the sample inspection has been completed, a hardness test shall be performed on 10 percent of the sample cutter bits and cut-off blades. If any of the test samples fail to meet the applicable hardness specified herein, the total lot from which the samples were selected shall be rejected.

4.6 Packaging inspection. Packaging shall be inspected for conformance with section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified in the contract or purchase order preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.1) preservation, packing, and marking shall be in accordance with PPP-P-40 level, as specified.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

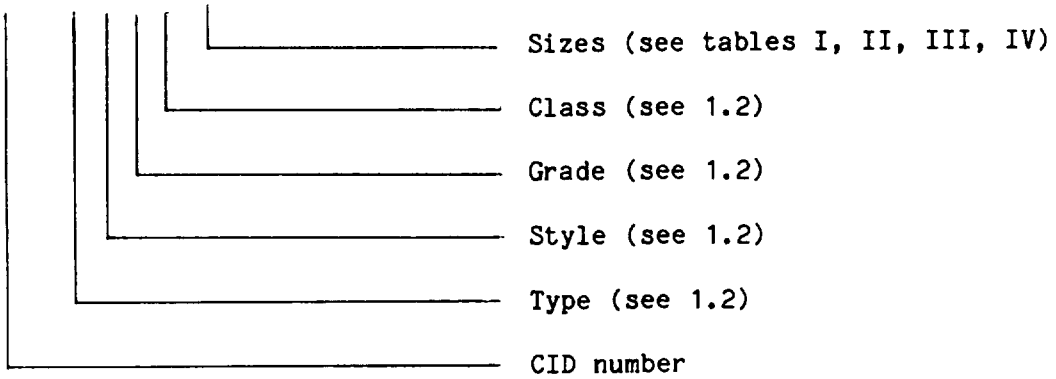
a. Title, number, and date of this commercial item description.
b. Type, style, grade, class, size and quantity required (see 1.2).
c. Packaging in accordance with PPP-P-40 if required: specify level (see 5.1).

6.2 Cross reference data. Tables I, II, III, and IV have been modified with sizes being added to the dimensions. This was done to make the tables compatible to the PIN numbers. Table V was condensed and modified to comply with quality requirements.

6.3 CID based part identification number. The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable Arabic numbers shall be substituted for Roman numerals.

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MILITARY INTEREST:

Custodians

Army - AL
Navy - SH
Air Force - 99

Review Activity

Air Force - 84

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

DLA - IP

Project 3455-0129

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE *(YYMMDD)*
3. DOCUMENT TITLE		
4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*		
5. REASON FOR RECOMMENDATION		

6. SUBMITTER		
a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON
c. ADDRESS *(Include Zip Code)*	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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